

累次极限存在 ~~且~~ 重极限存在

例: 1. $f(x, y) = \frac{xy}{x^2 + y^2}$

在 $(0, 0)$

累次极限存在且相等

重极限不存在

2. $g(x, y) = x \sin \frac{1}{y} + y \sin \frac{1}{x}$

在 $(0, 0)$

重极限为 0,

累次极限不存在.

证明重极限不存在:

存在两条路径, 沿着它们趋于 M_0 时, 极限不同
或
存在一条路径, 沿着它趋于 M_0 时, 极限不存在 $\Rightarrow$ 重极限不存在

求重极限:

- ① 化为一元函数 $f(x, y) = g(h(x, y))$
- ② 极坐标代换
- ③ 若已知重极限存在, 取一条路径即可.
- ④ 重极限. 累次极限都存在, 则相等 (9.1.16 推论)

补充习题 9.1.16 证明:

16. 证明: 当极限 $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = A$ 存在时,

(1) 若 $y \neq y_0$ 时, $\lim_{x \rightarrow x_0} f(x, y)$ 存在, 则 $\lim_{y \rightarrow y_0} \lim_{x \rightarrow x_0} f(x, y) = A$;

(2) 若 $x \neq x_0$ 时, $\lim_{y \rightarrow y_0} f(x, y)$ 存在, 则 $\lim_{x \rightarrow x_0} \lim_{y \rightarrow y_0} f(x, y) = A$.

只证 (1)

记 $\lim_{x \rightarrow x_0} f(x, y) = \phi(y) \quad (y \neq y_0)$

由 $\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y)$ 存在, $\forall \varepsilon > 0, \exists \delta > 0$, 当 $|x - x_0| < \delta, |y_1 - y_0| < \delta, |y_2 - y_0| < \delta$ 时,

$|f(x, y_1) - f(x, y_2)| < \varepsilon$ (Cauchy 收敛)

令 $x \rightarrow x_0$ 即有 $|\phi(y_1) - \phi(y_2)| < \varepsilon \quad \therefore \lim_{y \rightarrow y_0} \phi(y)$ 存在.

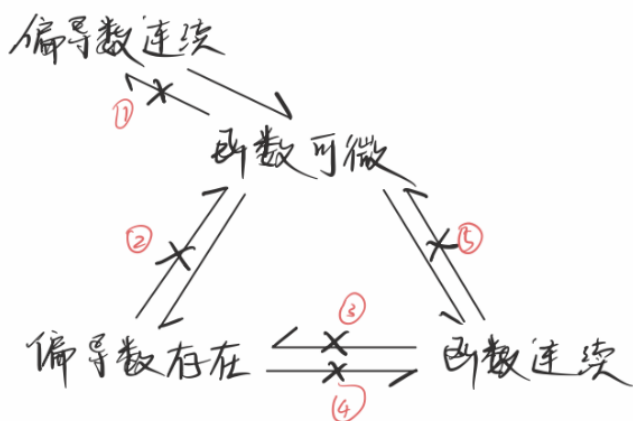
下证 $\lim_{y \rightarrow y_0} \phi(y) = A$

对上述 ε , $\exists \delta_1$, $0 < |x - x_0|$, $|y_1 - y_0| < \delta_1$ 时有
 $|f(x, y_1) - A| < \varepsilon$, $|f(x, y_1) - \phi(y_1)| < \varepsilon$

$$\therefore |\phi(y_1) - A| = |\phi(y_1) - f(x, y_1)| + |f(x, y_1) - A| < 2\varepsilon$$

$$\therefore \lim_{y \rightarrow y_0} \phi(y) = A$$

可微、连续、偏导存在的关系：



★反例：

① $f(x, y) = \begin{cases} (x^2 + y^2) \sin(\frac{1}{x^2 + y^2}), & x^2 + y^2 \neq 0 \\ 0, & (x, y) = (0, 0) \end{cases}$

② $f(x, y) = \frac{xy}{x^2 + y^2}$ 在 $(0, 0)$ 处

③ $f(x, y) = |xy|$ 在 $(0, 0)$ 处

④ $f(x, y) = \frac{xy}{x^2 + y^2}$ 在 $(0, 0)$

⑤ $f(x, y) = |x|$

隐函数 & 逆映射定理

1. 求导法：以 $\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$ 确定了 $u = u(x, y), v = v(x, y)$ 为例：

对 $\begin{cases} F=0 \\ G=0 \end{cases}$ 两边分别对 x, y 求偏导，

再从方程组中解出 $\frac{\partial u}{\partial x}, \frac{\partial v}{\partial x}, \frac{\partial u}{\partial y}, \frac{\partial v}{\partial y}$ 。

Remark. (Cramer 法则)

线性方程组 $\begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + \dots + a_{2n}x_n = b_2 \\ \vdots \\ a_{n1}x_1 + \dots + a_{nn}x_n = b_n \end{cases}$ 系数行列式 $D \equiv \begin{vmatrix} a_{11} & \dots & a_{1n} \\ a_{21} & \dots & a_{2n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{vmatrix}$

若 $D \neq 0$ ，则方程组有唯一解 $x_j = \frac{D_j}{D}$ ， $j=1, 2, \dots, n$ 。

其中 D_j 为系数矩阵 $\begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix}$ 将第 j 列替换为 $\begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$ 后所得的行列式。

2. 微分法：以 $\begin{cases} F(x, y, u, v) = 0 \\ G(x, y, u, v) = 0 \end{cases}$ 确定了 $u = u(x, y), v = v(x, y)$ 为例：

对 F, G 求微分得到 $\begin{cases} F'_x dx + F'_y dy + F'_u du + F'_v dv = 0 \\ G'_x dx + G'_y dy + G'_u du + G'_v dv = 0 \end{cases}$

解出 $\begin{cases} du = \frac{\partial u}{\partial x} dx + \frac{\partial u}{\partial y} dy \\ dv = \frac{\partial v}{\partial x} dx + \frac{\partial v}{\partial y} dy \end{cases}$ (红色的为待求解)

Rmk. 建议掌握方法，而不是背公式。

二元函数逆映射微商

$$\begin{pmatrix} dy_1 \\ dy_2 \end{pmatrix} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} \end{pmatrix} \begin{pmatrix} dx_1 \\ dx_2 \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} \end{pmatrix}^{-1} \begin{pmatrix} dy_1 \\ dy_2 \end{pmatrix} = \begin{pmatrix} dx_1 \\ dx_2 \end{pmatrix}$$

$$\therefore \begin{pmatrix} dx_1 \\ dx_2 \end{pmatrix} = \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{pmatrix} \begin{pmatrix} dy_1 \\ dy_2 \end{pmatrix}$$

$$\therefore \begin{pmatrix} \frac{\partial x_1}{\partial y_1} & \frac{\partial x_1}{\partial y_2} \\ \frac{\partial x_2}{\partial y_1} & \frac{\partial x_2}{\partial y_2} \end{pmatrix} = \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} \end{pmatrix}^{-1} = \frac{1}{|J|} \begin{pmatrix} \frac{\partial y_2}{\partial x_2} & -\frac{\partial y_1}{\partial x_2} \\ -\frac{\partial y_2}{\partial x_1} & \frac{\partial y_1}{\partial x_1} \end{pmatrix}$$

例. 设 $u = u(x, y)$, $v = v(x, y)$ 由方程组 $\begin{cases} u = f(u, v) \\ v = g(u, v) \end{cases}$ 确定.

求 $\frac{\partial u}{\partial x}$.

解: $\frac{\partial u}{\partial x} = \begin{vmatrix} u'_x & u'_y \\ v'_x & v'_y \end{vmatrix}$

$\begin{cases} u = f(u, v) \\ v = g(u, v) \end{cases}$ 两边对 x 求偏导

$\begin{cases} u'_x = f'_1 \cdot (u'_x \cdot x + u) + f'_2 \cdot (v'_x + 0) \\ v'_x = g'_1 \cdot (u'_x - 1) + g'_2 \cdot (2y v v'_x) \end{cases}$

$\Rightarrow \begin{cases} u'_x (1 - x f'_1) - v'_x f'_2 = u f'_1 \\ u'_x (-g'_1) + (1 - 2y v g'_2) v'_x = -g'_1 \end{cases}$

下用 Cramer 法则: $D = \begin{vmatrix} 1 - x f'_1 & -f'_2 \\ -g'_1 & 1 - 2y v g'_2 \end{vmatrix}$ $D_1 = \begin{vmatrix} u f'_1 & -f'_2 \\ -g'_1 & 1 - 2y v g'_2 \end{vmatrix}$ $D_2 = \begin{vmatrix} 1 - x f'_1 & u f'_1 \\ -g'_1 & -g'_1 \end{vmatrix}$

$u'_x = \frac{D_1}{D}$, $u'_y = \frac{D_2}{D}$.

再 $\begin{cases} u = f(u, v) \\ v = g(u, v) \end{cases}$ 两边同时对 y 求偏导 $\xrightarrow{\text{Cramer 法则}} u'_y, v'_y$.

补充: 齐次函数的欧拉定理 \ 逆定理

Def: 若 $f: D \subset \mathbb{R}^k \rightarrow \mathbb{R}$ 满足 $\forall a > 0$,

$f(ax_1, ax_2, \dots, ax_k) = a^n f(x_1, x_2, \dots, x_k)$,

则称 f 为 n 次齐次函数.

齐次函数欧拉定理:

若 $f(x_1, \dots, x_k)$ 为可微的 n 次齐次函数, 则有 $\sum_{i=1}^k \frac{\partial f}{\partial x_i} \cdot x_i = n f$.

证明: 由定义 $f(ax_1, ax_2, \dots, ax_k) = a^n f(x_1, x_2, \dots, x_k)$

两边对 a 求导得到:

$$\sum_{i=1}^k f'_i(ax_1, ax_2, \dots, ax_k) \cdot x_i = na^{n-1} f(x_1, \dots, x_k)$$

令 $a=1$ 即证.

逆定理: 满足 $\sum_{i=1}^k \frac{\partial f}{\partial x_i} \cdot x_i = nf$ 的可微函数 $f: D \subseteq \mathbb{R}^k \rightarrow \mathbb{R}$ 为 n 次齐次函数.

证明: 记 $x = (x_1, \dots, x_k)$, $g_x(a) = f(ax_1, ax_2, \dots, ax_k)$

$$\text{则) 有 } \frac{dg_x}{da} = \sum_{i=1}^k f'_i(ax_1, \dots, ax_k) \cdot x_i$$

$$\sum_{i=1}^k f'_i(x_1, \dots, x_k) \cdot x_i = nf(x_1, \dots, x_k)$$

$$\Rightarrow \sum_{i=1}^k f'_i(ax_1, \dots, ax_k) \cdot ax_i = nf(ax_1, \dots, ax_k)$$

$$\Rightarrow \sum_{i=1}^k f'_i(ax_1, \dots, ax_k) x_i = a^{-1} nf(ax_1, \dots, ax_k)$$

$$\Rightarrow \frac{dg_x(a)}{da} = a^{-1} n g_x(a)$$

$$\Rightarrow \frac{dg_x(a)}{g_x(a)} = n \cdot \frac{da}{a}$$

解微分方程 $g_x(a) = c \cdot a^n$, 令 $a=1$ 得 $c = g_x(1) = f(x_1, \dots, x_k)$

$$\text{即 } f(ax_1, \dots, ax_k) = f(x_1, \dots, x_k) \cdot a^n$$

应用: 若 $\mathbb{R}^2$ 上的可微函数满足 $xf'_x(x, y) + yf'_y(x, y) = 0$.

则 $f(x, y) \equiv \text{常数}$.

$$\text{proof: } \because x \cdot \frac{\partial f}{\partial x} + y \cdot \frac{\partial f}{\partial y} = 0 = 0 \cdot f(x, y)$$

由欧拉定理逆定理, $\forall t > 0$, 有

$$f(tx, ty) = t^0 f(x, y) = f(x, y)$$

$$\text{令 } t \rightarrow 0^+ \text{ 得 } f(x, y) = \lim_{(x, y) \rightarrow (0, 0)} f(tx, ty) \overset{\text{由可微}}{=} f(0, 0) = \text{const}$$

设 $f(x, y)$ 为开区域 $D \subseteq \mathbb{R}^2$ 上连续可偏导函数, u, v 为 $\mathbb{R}^2$ 上的夹角为 α 的单位向量, 证明:

$$\left(\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right) \sin^2 \alpha \leq 2 \left(\left(\frac{\partial f}{\partial u} \right)^2 + \left(\frac{\partial f}{\partial v} \right)^2 \right)$$

证明 1: 设方向 u 的方向角为 θ , 则 u 的方向余弦为 $\{\cos \theta, \sin \theta\}$

则: v 的方向余弦为 $\{\cos(\theta + \alpha), \sin(\theta + \alpha)\}$

$$\text{于是: } \frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \cos \theta + \frac{\partial f}{\partial y} \sin \theta, \quad \frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \cos(\theta + \alpha) + \frac{\partial f}{\partial y} \sin(\theta + \alpha)$$

$$\text{有: } \frac{\partial f}{\partial u} \sin(\theta + \alpha) = \frac{\partial f}{\partial x} \cos \theta \sin(\theta + \alpha) + \frac{\partial f}{\partial y} \sin \theta \sin(\theta + \alpha)$$

$$\text{和: } \frac{\partial f}{\partial v} \sin \theta = \frac{\partial f}{\partial x} \cos(\theta + \alpha) \sin \theta + \frac{\partial f}{\partial y} \sin(\theta + \alpha) \sin \theta.$$

$$\text{得: } \frac{\partial f}{\partial u} \sin(\theta + \alpha) - \frac{\partial f}{\partial v} \sin \theta = \frac{\partial f}{\partial x} \cos \theta \sin(\theta + \alpha) - \frac{\partial f}{\partial x} \cos(\theta + \alpha) \sin \theta$$

$$\text{即: } = \frac{\partial f}{\partial x} (\cos \theta \sin(\theta + \alpha) - \cos(\theta + \alpha) \sin \theta)$$

$$\text{即: } = \frac{\partial f}{\partial x} \sin \alpha.$$

设 $\sin \alpha \neq 0$ 时, (当 $\sin \alpha = 0$ 时不^等式成立)

$$\text{于是: } \frac{\partial f}{\partial x} = \frac{1}{\sin \alpha} \left(\frac{\partial f}{\partial u} \sin(\theta + \alpha) - \frac{\partial f}{\partial v} \sin \theta \right)$$

$$\text{同理: } \frac{\partial f}{\partial u} = \frac{\partial f}{\partial x} \cos \theta + \frac{\partial f}{\partial y} \sin \theta, \quad \frac{\partial f}{\partial v} = \frac{\partial f}{\partial x} \cos(\theta + \alpha) + \frac{\partial f}{\partial y} \sin(\theta + \alpha)$$

$$\text{有: } \frac{\partial f}{\partial u} \cos(\theta + \alpha) = \frac{\partial f}{\partial x} \cos \theta \cos(\theta + \alpha) + \frac{\partial f}{\partial y} \sin \theta \cos(\theta + \alpha)$$

$$\text{和: } \frac{\partial f}{\partial v} \cos \theta = \frac{\partial f}{\partial x} \cos(\theta + \alpha) \cos \theta + \frac{\partial f}{\partial y} \sin(\theta + \alpha) \cos \theta.$$

$$\text{得: } \frac{\partial f}{\partial u} \cos(\theta + \alpha) - \frac{\partial f}{\partial v} \cos \theta = \frac{\partial f}{\partial y} \sin \theta \cos(\theta + \alpha) - \frac{\partial f}{\partial y} \sin(\theta + \alpha) \cos \theta$$

$$\text{即: } = \frac{\partial f}{\partial y} (\sin \theta \cos(\theta + \alpha) - \sin(\theta + \alpha) \cos \theta)$$

$$\text{即: } = -\frac{\partial f}{\partial y} \sin \alpha.$$

$$\text{是: } \frac{\partial f}{\partial x} = \frac{1}{\sin \alpha} \left(\frac{\partial f}{\partial u} \sin(\theta + \alpha) - \frac{\partial f}{\partial v} \sin \theta \right)$$

$$\text{和: } \frac{\partial f}{\partial y} = -\frac{1}{\sin \alpha} \left(\frac{\partial f}{\partial u} \cos(\theta + \alpha) - \frac{\partial f}{\partial v} \cos \theta \right)$$

$$\text{故: } \left(\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right) = \frac{1}{\sin^2 \alpha} \left(\left(\frac{\partial f}{\partial u} \right)^2 + \left(\frac{\partial f}{\partial v} \right)^2 - 2 \frac{\partial f}{\partial u} \frac{\partial f}{\partial v} \cos \alpha \right)$$

$$\text{即: } \leq \frac{1}{\sin^2 \alpha} \left(\left(\frac{\partial f}{\partial u} \right)^2 + \left(\frac{\partial f}{\partial v} \right)^2 + 2 \left| \frac{\partial f}{\partial u} \right| \left| \frac{\partial f}{\partial v} \right| \right)$$

$$\text{即: } \leq \frac{2}{\sin^2 \alpha} \left(\left(\frac{\partial f}{\partial u} \right)^2 + \left(\frac{\partial f}{\partial v} \right)^2 \right)$$

$$\text{于是: } \left(\left(\frac{\partial f}{\partial x} \right)^2 + \left(\frac{\partial f}{\partial y} \right)^2 \right) \sin^2 \alpha \leq 2 \left(\left(\frac{\partial f}{\partial u} \right)^2 + \left(\frac{\partial f}{\partial v} \right)^2 \right).$$